Outcome of Extratemporal Epilepsy Surgery Experience of a Single Center

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OBJECTIVE: Our aim was to determine the surgical outcome in adult patients with intractable extratemporal epilepsy and follow it over time.

METHODS: We retrospectively studied the operative outcome in 218 consecutive adult patients with extratemporal lesions who underwent resective surgical treatment for intractable partial epilepsy in the Bethel Epilepsy Center, Bielefeld, Germany, between 1991 and 2005. Patients were divided into three groups according to the 5-year period in which the surgical procedure took place.

RESULTS: Group I (1991–1995) consisted of 64 patients. The postoperative Engel Class I outcome was 50% at 6 months, 44.4% at 2 years, and 45.2% at 5 years. Group II (1996–2000) included 91 patients. Engel Class I outcome was 57.1% at 6 months, 53.8% at 2 years, and 53.5% at 5 years. In Group III (2001–2005), there were 63 patients. Engel Class I outcome was 65.1% at 6 months, 61.3% at 2 years, and 60.6% at 5 years. Short duration of epilepsy, surgery before 30 years of age, pathological findings of neoplasm, and well-circumscribed lesions on the preoperative magnetic resonance imaging scan were good prognostic factors. Poor prognostic factors were one or more of the following: psychic aura, generalized tonic-clonic seizure, versive seizure, history of previous surgery, and focal cortical dysplasia. On multivariate analysis, only the presence of well-circumscribed lesions on preoperative magnetic resonance imaging predicted a positive outcome ($P = 0.001$).

CONCLUSION: Our results indicate that extratemporal epilepsy surgery at the Bethel Epilepsy Center has become more effective in the treatment of extratemporal epilepsy patients over the years, ensuring continuous improvement in outcome. This improvement can be attributed mainly to more restrictive patient selection.

KEY WORDS: Epilepsy surgery, Extratemporal, Long-term outcome

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Epilepsy surgery is now widely accepted for the management of medically intractable localization-related epilepsy (34). Although results of extratemporal resection in adult patients with focal epilepsy have been reported previously, many of the studies included only a small number of patients and took place before magnetic resonance imaging (MRI) was widely implemented. Most reports include patients with both extratemporal and temporal lobe epilepsy of diverse causes (33, 36, 40). Results reported at the Second International Conference on Epilepsy Surgery suggest that the outcome of seizure control has improved in patients with temporal lobe resections but is only stable in patients with extratemporal resections (10, 33). Recent developments in epilepsy surgery have increased the need for more follow-up studies (37). Whereas several studies have estimated the outcome of extratemporal epilepsy surgery, little attention has been paid to any changes in the extratemporal outcome over time. This report addresses the following questions: Has extratemporal epilepsy surgery outcome altered over time? What is our current state of extratemporal epilepsy surgery outcome? What are the characteristics of patients with the best outcome? What variables correlate with good long-term outcome? Our study includes only adult patients with extratemporal epilepsy who

ABBREVIATION: CI, confidence interval; **MRI**, magnetic resonance imaging had surgery in an era that could benefit from MRI. They were under the care of the same neurologists and surgeons throughout the time periods in question. Although previous studies found that the response to epilepsy surgery during the first year of follow-up is a reliable indicator of long-term outcome (8, 41), we followed the progress of most of the patients for 5 years (4.1 ± 1.5 yr) postoperatively.

PATIENTS AND METHODS

We retrospectively reviewed the records of all adult patients (≥16 yr of age) who underwent epilepsy surgery at the Bethel Epilepsy Center, Bielefeld, Germany, from 1991 to 2005. There were 243 consecutive patients. We excluded patients who only underwent biopsy, patients with Rasmussen's encephalitis (a progressive disease with special criteria), patients who did not have electroencephalographic video monitoring, and patients who had unclear extratemporal resections. Nine patients had normal brain tissue in resected pathological specimens. Our study includes 218 patients divided into three groups according to the time period of operation: Group I patients were operated on from 1991 through 1995, Group II patients were operated on from 1996 through 2000, and Group III patients were operated on from 2001 through 2005.

Patient Characteristics

Group I consisted of 64 patients (29.3%); 41 patients were male (64%), and 23 were female (36%). The mean age at epilepsy onset was 11.7 ± 10.8 years (range, 0.08–51.0 yr; 95% confidence interval [CI], 8.8–15.0 yr). The mean age at surgery was 29.3 ± 11.3 years (range, 16–57 yr; 95% CI, 25–32 yr). The mean duration of epilepsy was 17.5 ± 9.8 years (range, 1–38 yr; 95% CI, 14.2–19.6 yr); and the mean duration of follow-up was 4.9 ± 0.5 years (range, 1–5 yr).

Group II was composed of 91 patients (41.7%); 50 patients were male (55%), and 41 were female (45%). The mean age at onset of epilepsy was 13.2 ± 9.3 years (range, 1–52 yr; 95% CI, 10.9–14.6 yr). The mean age at surgery was 28.1 ± 10.1 years (range, 16–59 yr; 95% CI, 26.1–30.4 yr). The mean duration of epilepsy was 14.9 ± 9.0 years (range, 0.83–44 yr; 95% CI, 13.2–17.0 yr); and the mean duration of follow-up was 4.9 ± 0.5 years (range, 2–5 yr).

Group III included 63 patients (29%); 38 patients were male (60%), and 25 were female (40%). The mean age at onset was 17.4 ± 15.9 years (range, 0.04–65.0 yr; 95% CI, 13.6–21.2 yr). The mean age at surgery was 32.2 ± 13.7 years (range, 16.0–69.0 yr; 95% CI, 29.1–35.6 yr). The mean duration of epilepsy was 13.8 ± 13.8 years (range, 0.91–65.4 yr; 95% CI, 11.4–17.9 yr); and the mean duration of follow-up was 2.5 ± 1.3 years (range, 1–5 yr).

The clinical characteristics of these patients, as well as detailed differences among the groups, are summarized in *Table 1*. The foregoing values are given as the mean number of years ± standard deviation.

Preoperative Evaluation

Preoperative workup for epilepsy surgery candidates includes recording the patient's history and carrying out a physical examination, neuropsychological evaluation, visual field test, and MRI examination, as well as inpatient video electroencephalographic monitoring and invasive monitoring with some patients for delineation of eloquent cortical areas. In our study, 75 patients (34.4%) underwent invasive monitoring in the form of subdural grids: 38 (59.3%) in Group 1, 30 (32.9%) in Group II, and 7 (11.1%) in Group III.

Neuroimaging Modalities

Until 1998, a 1-T MRI scanner was used, and later a 1.5-T MRI scanner was available. A specific protocol for epilepsy patients was followed, using functional MRI, Wada test, positron emission tomography, and single-photon emission computed tomography (interictal and ictal) for the majority of the patients and additionally conducting MRI angiography in a few cases.

Surgical Procedure and Postoperative Evaluation

All patients underwent resective surgery in the form of lesionectomy, cortical resection, or lobectomy. There were a total of 34 patients (16.3%) who had surgery in the central area: 12 patients (18.7%) from Group I, 12 (13.1%) from the Group II, and 10 (15.8%) from Group III. Four patients (11.7%) had incomplete resections because of overlapping of the motor area and underwent combined lesionectomy with multiple subpial transections.

The same surgical team carried out all operations. Subdural grid or strip electrodes were used for invasive monitoring. Intraoperative electrocorticography was routinely used, except for patients with extraoperative invasive monitoring. Postoperative inpatient evaluation routinely included MRI and electroencephalography after 6 months and 2 years.

Data Collection

Research data in this study were drawn from four main sources: the patient's medical records, the postoperative routine follow-up program, additional outpatient appointments, and telephone and mail questionnaires. Families were consulted in many cases, especially for patients with mental retardation or with psychological problems. In 10 patients (4.5%), the data were missing after 2 years' follow-up. Another five patients (2.25%) died: one (0.4%) died in status epilepticus, and four (1.8%) died as a result of other medical causes. Histopathological information was extracted from histopathological reports in the patients' files. In our analysis, there were pathological findings in the resected specimen in 209 cases (95.9%), and no pathological findings in 9 cases (4.1%).

Outcome Evaluation

Seizure outcome was classified by using the modified seizure classification of Engel. Engel Class I patients were seizure-free; or they had only nondisabling, simple, partial seizures; or they had some disabling seizures after surgery but had been free of disabling seizures for at least 2 years; or generalized convulsions occurred only after withdrawal of antiepileptic drugs. Engel Class II patients rarely had seizures (>85% reduction). Engel Class III patients experienced worthwhile improvement, with a greater than 50% reduction in seizure frequency. Engel Class IV patients experienced no significant reduction in seizure frequency.

Outcome in relation to preoperative MRI pathological findings was correlated with well-circumscribed lesions (neoplastic, vascular, and cystic lesions) or less-well-circumscribed lesions (cortical dysplasia, malformation of cortical development, and gliosis) based on the MRI shape of the lesions. For all patients, preoperative MRI was reevaluated visually in the case conference by two epileptologists and an epilepsy neurosurgeon to determine whether the appearance of a lesion was well circumscribed or less well circumscribed.

Statistical Analysis

Categorical data of patients were compared by using the two-tailed Fisher's exact test. Mean numerical variables were compared using the Student's independent t test. The χ^2 test and the rank-sum test were used for univariate analyses. We additionally examined the independent effect of the variables as possible predictors of surgical outcome

TABLE 1. Characteristics of 218 patients who underwent surgery over a 15-year period[a]

Variable	Group I (%)	Group II (%)	Group III (%)	Total (%)
Mean age at onset (yr)	11.7 ± 10.8	13.2 ± 9.3	17.4 ± 15.9	14.0 ± 12.1
Mean age at surgery (yr)	29.3 ± 11.3	28.1 ± 10.1	32.2 ± 13.7	29.6 ± 11.7
Mean age at second surgery (yr)	28.0 ± 9.7	32.3 ± 10.1	29.0 ± 14.1	30.3 ± 10.0
Mean duration of epilepsy (yr)	17.5 ± 9.8	14.9 ± 9.0	13.8 ± 13.7	15.7 ± 10.8
Mean duration of follow-up (yr)	4.9 ± 0.5	4.9 ± 0.5	2.4 ± 1.1	4.2 ± 1.4
Febrile convulsion	2 (3.1)	3 (3.2)	2 (3.1)	7 (3.2)
Family history	1 (1.5)	3 (3.2)	3 (4.3)	7 (3.2)
Head trauma	4 (6.2)	7 (7.6)	3 (4.3)	14 (6.4)
Previous surgery	14 (21.8)	25 (27.4)	17 (26.9)	56 (25.6)
Prenatal infarction	3 (4.6)	8 (8.7)	4 (6.3)	15 (6.8)
MCD	5 (7.8)	5 (5.4)	3 (4.7)	13 (5.9)
CNS infections	1 (1.5)	5 (5.4)	4 (6.3)	10 (4.5)
Aura	50 (78.1)	66 (72.5)	43 (68.2)	159 (72.9)
Unclassified seizure	2 (3.1)	4 (4.3)		6 (2.7)
Simple motor seizure	7 (10.9)	5 (5.4)	5 (7.9)	17 (7.7)
Complex partial seizure	34 (53.1)	43 (47.2)	18 (28.5)	95 (43.6)
Generalized seizure	15 (23.4)	23 (25.2)	31 (49.2)	69 (31.6)
Combined seizure form	8 (12.5)	14 (15.3)	9 (14.2)	31 (14.2)
Pathological finding				
Neoplasm	22 (34.4)	29 (31.90)	28 (44.4)	79 (36.2)
FCD	24 (37.5)	30 (33.0)	14 (22.2)	68 (31.2)
Vascular	6(9.4)	6 (6.6)	12 (19.0)	24 (11.0)
Gliosis	4 (6.3)	14 (15.40)	3 (4.8)	21 (9.6)
No abnormality	5 (7.8)	4 (4.4)		9 (4.1)
Cystic lesion	1 (1.6)	2 (2.2)	3 (4.8)	6 (2.8)
MCD		2 (2.2)	1 (1.6)	3 (1.4)
Sclerosis	1 (1.6)	1 (1.1)	1 (1.6)	3 (1.4)
Atrophy		2 (2.2)	1 (1.6)	3 (1.4)
Inflammation	1 (1.6)	1 (1.1)		2 (0.9)

[a] MCD, malformation of cortical development; CNS, central nervous system; FCD, focal cortical dysplasia.

through binary logistic regression analysis by using the SPSS statistical program (Version 11.0; SPSS, Inc., Chicago, IL). The magnitude of the association between selected variables and surgical outcome was measured by the crude and adjusted odds ratio. Results were considered significant for values of $P \leq 0.05$. Cox multivariate stepwise regression was used to evaluate the predictors for Engel Class I outcome.

RESULTS

Overall Outcome

Engel Class I outcome was found in 125 patients (57.3%) at the 6-month follow-up, in 121 patients (56.0%) at the 1-year follow-up, in 115 patients (53.2%) at the 2-year follow-up, and in 94 patients (51.9%) at the 5-year follow-up. The statistical difference among the groups with regard to Engel Class I outcome was $P = 0.375$ at 6 months, $P = 0.125$ at 1 year, $P = 0.03$ at 2 years, and $P = 0.230$ at 5 years. *Figure 1* shows the percentage of Engel Class I over time in all groups. Numbers and percentages of different Engel classes in all groups are summarized in *Table 2*.

The outcome for Group I was as follows: 32 patients (50.0%) were in Engel Class I at the 6-month follow-up, 31 patients (49.2%) at 1 year, 28 patients (45.2%) at 2 years, and 28 patients (44.4%) at 5 years. After 5 years of postsurgical follow-up, 77.4% of the patients had a favorable outcome (Classes I–III), and 22.6% had no significant changes (Class IV).

In Group II, an Engel Class I outcome was found in 52 patients (57.1%) at the 6-month follow-up, in 51 patients (56.0%) at 1 year, in 49 patients (53.8%) at 2 years, and in 46 patients (53.5%) at 5 years. After 5 years of postsurgical follow-

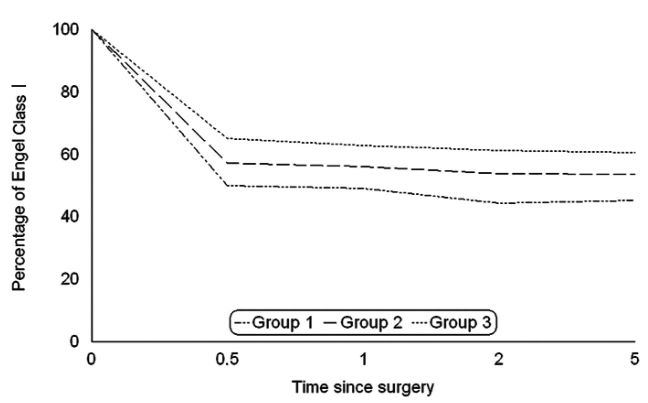

FIGURE 1. *Graph showing the percentage of patients in the three groups in Engel Class I at different time points. The line representing Group III also represents the current state of the outcome in extratemporal epilepsy surgery.*

up, 87.2% of the patients had a favorable outcome, and 12.8% had no changes.

For Group III, an Engel Class I outcome was found in 41 patients (65.1%) at the 6-month follow-up, in 39 patients (62.9%) at 1 year, in 38 patients (61.3%) at 2 years, and in 20 patients (60.6%) at 5 years. At the 5-year follow-up, 90.9% of patients had a favorable outcome, and 9.1% had no significant postsurgical changes.

TABLE 2. Numbers and percentages of patients in the different Engel classes at 6 months and 1, 2, and 5 years

Engel Class	Group I (%)	Group II (%)	Group III (%)	Total (%)
At 6 mo				
I	32 (50.0)	52 (57.1)	41 (65.1)	125 (57.3)
II	9 (14.1)	10 (11.0)	6 (9.5)	25 (11.5)
III	7 (10.9)	14 (15.4)	7 (11.1)	28 (12.8)
IV	16 (25.0)	15 (16.5)	9 (14.3)	40 (18.3)
At 1 yr				
I	31 (49.2)	51 (56.0)	39 (62.9)	121 (56.0)
II	10 (15.9)	11 (12.1)	9 (14.5)	30 (13.9)
III	8 (12.7)	15 (16.5)	6 (9.7)	29 (13.4)
IV	14 (22.2)	14 (15.4)	8 (12.9)	36 (16.7)
At 2 yr				
I	28 (44.4)	49 (53.8)	38 (61.3)	115 (53.2)
II	17 (27.0)	12 (13.2)	10 (16.1)	39 (18.1)
III	7 (11.1)	16 (17.6)	7 (11.3)	30 (13.9)
IV	11 (17.5)	14 (15.4)	7 (11.3)	32 (14.8)
At 5 yr				
I	28 (45.2)	46 (53.5)	20 (60.6)	94 (51.9)
II	10 (16.1)	10 (11.6)	5 (15.2)	25 (13.8)
III	10 (16.1)	19 (22.1)	5 (15.2)	34 (18.8)
IV	14 (22.6)	11 (12.8)	3 (9.1)	28 (15.5)

Outcome in Relation to Resection Location

Frontal Lobe

In Group I, 25 patients (39.1%) had frontal lobe resections. Engel Class I outcome was found in 13 patients (52.0%) at the 6-month postoperative follow-up, in 13 (52.0%) at 1 year, in 11 (44.0%) at 2 years, and in 11 (44.0%) at 5 years. At the 5-year follow-up, 68% of patients had a favorable outcome (Classes I–III), and 32% had no significant changes (Class IV) (*Fig. 2*).

In Group II, there were 37 patients (40.7%) with frontal lobe resections. Engel Class I outcome was found in 21 patients (56.80%) at 6 months, in 21 patients (56.80%) at 1 year, in 19 patients (51.40%) at 2 years, and in 18 patients (54.50%) at 5 years. At the 5-year follow-up, 93.9% of patients had a favorable outcome, and 6.1% had no significant postsurgical changes.

In Group III, 33 patients (52.4%) had frontal lobe resections. Engel Class I outcome was found in 23 patients (69.7%) at 6 months, in 22 patients (66.7%) at 1 year, in 21 patients (63.7%) at 2 years, and in 11 patients (61.1%) at 5 years. At the 5-year follow-up, 88.2% of patients had a favorable outcome, and 11.8% had no significant changes.

The outcome in patients who were operated on in the motor area was slightly less favorable than the outcome for those operated on in other areas of the extratemporal cortex. Engel Class I was found in 17 patients (50%) at the 6-month follow-up, in 16 patients (47.1%) at 2 years, and in 11 patients (45.8%) at 5 years.

Posterior Cortical Resections

In Group I, 31 patients (48.4%) had posterior cortical epilepsy resections. Engel Class I outcome was found in 18 patients (58.1%) at the 6-month postoperative follow-up, in 17 patients (56.7%) at 1 year, in 16 patients (53.3%) at 2 years, and in 15

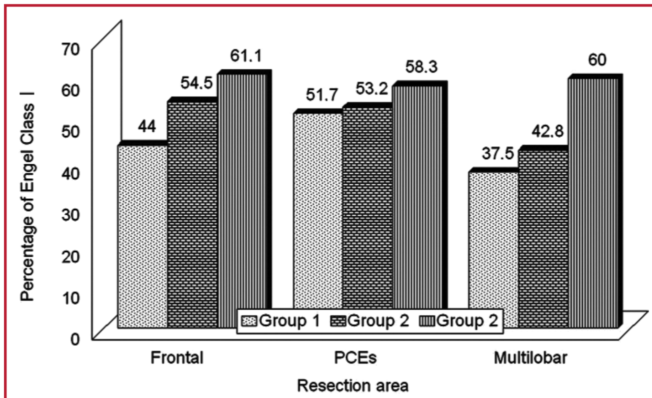

FIGURE 2. *Bar graph showing the percentage of patients in the three groups in Engel Class I at the 5-year follow-up, categorized on the basis of the type of resection area.* PCE, *posterior cortical epilepsy.*

patients (51.7%) at 5 years. At the 5-year follow-up, 89.7% of these patients had a favorable outcome, and 10.3% had no significant postsurgical changes.

In Group II, 47 patients (51.6%) underwent resections of the posterior cortex. Engel Class I outcome was found in 27 patients (57.4%) at 6 months, in 27 patients (57.4%) at 1 year, in 26 patients (55.3%) at 2 years, and in 25 patients (53.2%) at 5 years. At the 5-year follow-up examination, 85.1% of patients had a favorable outcome, and 14.9% had no significant postsurgical changes.

In Group III, 25 patients (39.7%) had posterior cortical resections. Engel Class I outcome was found in 15 patients (60.0%) at the 6-month postoperative follow-up, in 15 patients (60.0%) at 1 year, in 14 patients (58.3%) at 2 years, and in 7 patients (58.3%) at 5 years. At the 5-year follow-up, 81.7% of patients had a favorable outcome, and 8.3% had no significant postsurgical changes.

We had 11 patients with purely parietal lobe resections and 6 patients with purely occipital lobe resections. Engel Class I was found in 8 patients (72.7%) at the 2-year follow-up and in 6 patients (70%) at the 5-year follow-up in the parietal group. In patients who had purely occipital resections, 4 (66.7%) were in Engel Class I at the 2- and 5-year follow-ups.

Multilobar Resections

In Group I, 8 patients (12.5%) had multilobar resections. At 2 and 5 years, 3 patients (37.5%) were in Engel Class I. In Group II, 7 patients (7.7%) underwent multilobar resections. Engel Class I outcome was found in 3 patients (42.8%) at 2 and 5 years. In Group III, 5 patients (7.9%) had multilobar resections. Engel Class I outcome was found in 3 patients (60.0%) at 2 and 5 years. At the 5-year follow-up, 88.3% of the patients in Group I, 87.5% of the patients in Group II, and 88.3% of the patients in Group III had a favorable outcome.

Outcome in Relation to the Preoperative MRI Pathological Findings

In Group I, 31 patients (48.4%) had well-circumscribed lesions, and 33 patients (51.6%) had less-well-circumscribed lesions. Engel Class I outcome for the patients with well-circumscribed lesions after 6 months and 1, 2, and 5 years was found in 17 patients (54.8%), 16 patients (51.6%), 15 patients (50%), and 14 patients (48.3%), respectively. Engel Class I outcome for the patients with less-well-circumscribed lesions after 6 months and 1, 2, and 5 years was found in 15 patients (45.5%), 14 patients (42.4%), 13 patients (39.3%), and 13 patients (39.3%), respectively. At the 5-year follow-up, 86.7% of the patients with well-circumscribed lesions and 77.1% of those with less-well-circumscribed lesions had a favorable outcome.

In Group II, 37 patients (40.7%) had well-circumscribed lesions, and 54 patients (59.3%) had less-well-circumscribed lesions. Engel Class I outcome for the well-circumscribed lesions after 6 months and 1, 2, and 5 years was found in 25 patients (67.6%), 24 patients (64.9%), 23 patients (62.2%), and 22 patients (61.1%), respectively. Engel Class I outcome for the patients with less-well-circumscribed lesions after 6 months and 1, 2, and 5 years was found in 31 patients (57.4%), 29 patients (53.7%), 24 patients (44.4%), and 25 patients (50.0%), respectively. At the 5-year follow-up, 83% of the patients with well-circumscribed lesions and 80% of those with less-well-circumscribed lesions had a favorable outcome.

In Group III, 42 patients (66.7%) had well-circumscribed lesions, and 21 patients (33.3%) had less-well-circumscribed lesions. Engel Class I outcome for those patients with well-circumscribed lesions after 6 months and 1, 2, and 5 years was found in 32 patients (76.2%), 31 patients (73.8%), 30 patients (71.4%), and 17 patients (70.8%), respectively. Engel Class I outcome for patients with less-well-circumscribed lesions after 6 months and 1, 2, and 5 years was found in 12 patients (57.1%), 11 patients (55.0%), 10 patients (50%), and 5 patients (55.6%), respectively. At the 5-year follow-up, 100% of the patients with well-circumscribed lesions and 90% of those with less-well-circumscribed lesions had a favorable outcome.

Engel Class I outcome showed significant differences among patients with regard to well-circumscribed and less-circumscribed lesions on MRI ($P = 0.03$) (*Fig. 3*).

Outcome According to Pathological Findings

At the 2-year follow-up, Engel Class I was found in 50 patients (64.1%) with neoplasms, in 15 patients (62.5%) with vascular lesions, in 25 patients (37.3%) with focal cortical dysplasia, in 10 patients (47.6%) with gliosis, and in 4 patients (44.4%) with no pathological lesions (*Fig. 4*). The numbers and percentages of the patients in Engel Class I at the 6-month and 1-, 2-, and 5-year follow-ups are summarized in *Table 3*. Although there was a significant difference at the first-year follow-up ($P = 0.012$) between the lesional and nonlesional groups and an apparent improvement in the percentage of Engel Class I over time, this result is considered not conclusive because of the small number of patients in the nonlesional group.

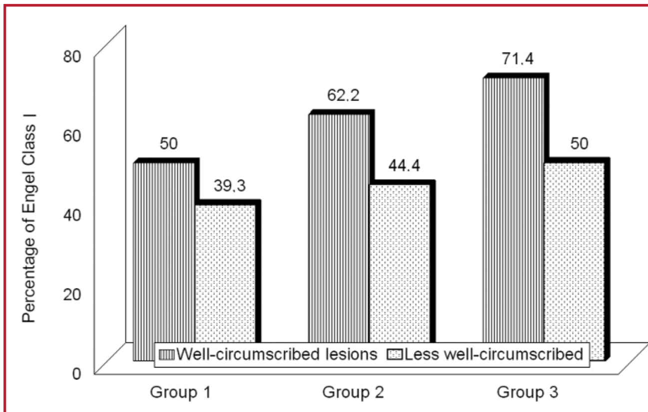

FIGURE 3. *Bar graph showing the percentage of patients in the three groups in Engel Class I at the 2-year follow-up, categorized on the basis of the type of lesion on preoperative magnetic resonance imaging (MRI) (well-circumscribed or less-well-circumscribed).*

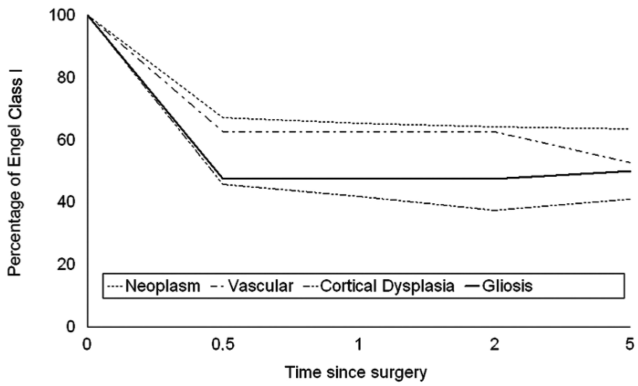

FIGURE 4. *Line graph showing the percentage of patients in Engel Class I over time in the entire sample, categorized on the basis of different types of abnormality.*

TABLE 3. Numbers and percentages of patients in Engel Class I at 6 months and 1, 2, and 5 years in most frequent pathological categories in the entire sample

Abnormality type	6 mo (%)	1 yr (%)	2 yr (%)	5 yr (%)
Neoplasm	53 (67.1)	51 (65.4)	50 (64.1)	40 (63.5)
Cortical dysplasia	31 (45.6)	28 (41.8)	25 (37.3)	25 (41.0)
Vascular	15 (62.5)	15 (62.5)	15 (62.5)	10 (52.6)
Gliosis	10 (47.6)	10 (47.6)	10 (47.6)	8 (50.0)
No abnormality	3 (33.3)	3 (33.3)	4 (44.4)	4 (44.4)

Prognostic Factors (Univariate and Multivariate Analysis)

In our study, the univariate analysis showed (*Table 4*) that there was a significant correlation of good outcome with each of the following findings: a well-circumscribed lesion on the preoperative MRI scan (*Fig. 5*), short duration of epilepsy (*Fig. 6*), early surgical interference, and a neoplasm in the resected specimen. There was a significant correlation of poor outcome with each of the following findings: a psychic aura, versive seizures, tonic-clonic seizures, history of previous surgery, and focal cortical dysplasia in the resected specimen. *Table 4* lists the results of the univariate analysis in the entire sample.

In the Cox multivariate stepwise regression analysis, only the presence of a well-circumscribed lesion on the preoperative MRI scan could predict the long- and short-term outcome (P = 0.001; hazard ratio, 0.523; 95% CI, 0.361–0.756). All of the other factors that correlated with the outcome in the univariate analysis were unable to provide predictive value for the outcome in the Cox multivariate stepwise regression analysis.

Postoperative Complications

In Group I, 8 patients (13.5%) had various types of complications; 5 patients (8.4%) had permanent disabilities, and 3

TABLE 4. Univariate analysis for significant variables[a]

Variables	*P* value	HR	95% CI
Well-circumscribed lesion on MRI	0.03	0.658	0.489–0.885
Invasive monitoring	0.0360	1.78	1.01–3.12
Psychic aura	0.0100	1.96	1.34–2.86
Versive seizure	0.012	2.870	1.261–6.532
Tonic-clonic seizures	0.024	2.56	0.05–6.19
History of previous surgery	0.045	2.033	1.017–4.061
FCD in abnormality	0.021	2.980	1.180–7.526
Neoplasm in abnormality	0.031	1.54	1.06–2.23
Duration of epilepsy (yr)			
≤5 (n = 37) (reference)			
>5 (n = 181)	0.044	0.631	0.404–0.987
Age at surgery (yr)			
≤30 (n = 119) (reference)			
>30 (n = 99)	0.042	0.349	0.127–0.964

[a] HR, hazard ratio; CI, confidence interval; MRI, magnetic resonance imaging; FCD, focal cortical dysplasia.

patients (5%) had transient morbidity. In Group II, 9 patients (10.3%) had complications; 4 patients (4.5%) had permanent disabilities, and 5 patients (5.5%) had transient morbidity. In Group III, 4 patients (6.3%) had complications; 1 patient (1.5%) had a permanent disability, and 3 patients (4.7%) had transient morbidity. *Figure 7* shows the type of complications and the number of patients in the entire sample.

DISCUSSION

A considerable amount of literature has been published on the outcome of extratemporal epilepsy surgery. These studies showed a great difference in the outcome attributable to different types of lesions, different types of surgery, and the absence of a standard surgical procedure for extratemporal epilepsy surgery compared with temporal lobe epilepsy surgery. Unlike most researchers, we focused on the changes of extratemporal epilepsy surgery outcome over time and factors that play a role in causing these changes. Our patients were divided into three groups on the basis of their year of surgery to detect outcome changes over time and changes in the characteristics of the patients who were selected for surgery. The patients were followed for 5 years postoperatively. Engel Class I outcome progressed between 1991 and 2005 from 45.2% to 61.3% at the 2-year follow-up and from 44.4% to 60.6% at the 5-year follow-up. This change in the outcome is statistically significant at the 2-year follow-up (P = 0.03).

Published results show various outcome figures: Zentner et al. (43) reported 54% in Class I, Mathern et al. (20) reported 46%, Wyllie et al. (42) reported 35%, and Spencer et al. (35) reported 56% in a multicenter study. Generally, the reports pub-

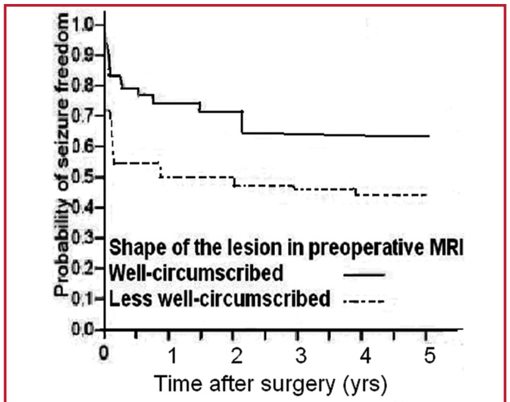

FIGURE 5. *Kaplan-Meier graph showing the proportion of patients in Engel Class I, categorized on the basis of shape of the lesion on the preoperative magnetic resonance imaging scan.*

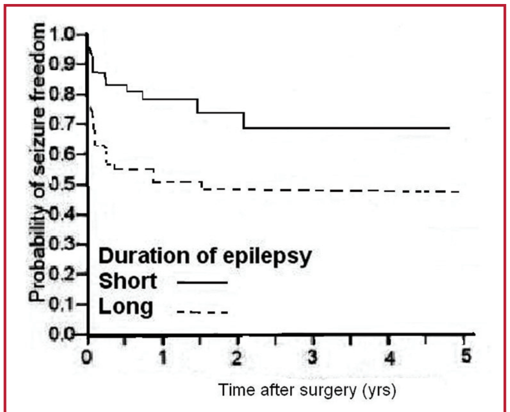

FIGURE 6. *Kaplan-Meier graph showing the proportion of patients in Engel Class I, categorized on the basis of preoperative duration of epilepsy.*

lished over the last 15 years have varied from 35 to 80% (2, 11, 14, 34, 35, 40, 44).

From the previous reports, it is not clear whether the outcome of extratemporal epilepsy surgery has progressed over time. At the Second International Conference on Epilepsy Surgery, it was suggested that outcomes for extratemporal epilepsy surgery had not changed since the time of the first Congress 6 years previously. This report (12, 46) from 1993 included patients operated on before the era of high-resolution MRI and the advancement of operative techniques. Tellez-Zenteno et al. (37) reported that outcome in the studies included combined temporal and extratemporal resections that improved more readily than purely temporal resections because of the improvement in outcome of the extratemporal resections. Our results are in line with studies showing more improvement in outcome of extratemporal resections.

Frontal Lobe Surgery Outcome

Previous reports showed great variation in the outcome after frontal lobe surgery: larger studies reported 50 to 55% favorable outcomes (11, 32, 38). However, seizure-free outcome was reported to reach 80% for specific subgroups of frontal lobe epilepsy surgery (12, 15, 19, 32, 44). Our results show a progressive increase of Engel Class I over time; Engel Class I had increased from 44% to 63.7% by the 2-year postoperative follow-up. By the 5-year follow-up, Engel Class I had increased from 44% to 61.1%. Some authors reported that frontal lobe epilepsy surgery improved over time (15, 16, 28). Our results are in accordance with previous reports and support this improvement of frontal lobe epilepsy surgery outcome.

Posterior Cortical Epilepsy Outcome

Our results show a continuous improvement of Engel Class I in posterior cortical epilepsies over time. Engel Class I increased from 53.3% to 55.3% after 2 years and from 51.7% to 58.3% after 5 years. In pure parietal lobe resection, Engel Class I reached 72.2% and 66.7%, respectively, over the same time periods in pure occipital lobe surgery.

Posterior cortical epilepsies in previous series range from 40 to 80% (2, 16, 19, 43). Recently, Kun Lee et al. (18) reported that 61.5% of patients were seizure-free after occipital lobe surgery. Dalmagro et al. (9) reported a 65.11% seizure-free rate, a percentage that is a little higher than our results. The reason for this difference may be that 20.6% of our patients had a previous history of surgery, a factor found to be associated with poor outcome in our study.

Multilobar Epilepsy Surgery Outcome

Multilobar epilepsy surgery showed some improvement despite the fact that this group had the least favorable outcomes in our study. Engel Class I changed from 37.5% to 60% at 2 and 5 years postoperatively. Devinsky et al. (10) reported 31% Class I and 23% Class II after multilobar epilepsy surgery, including operations on the eloquent cortex. However, the outcomes after

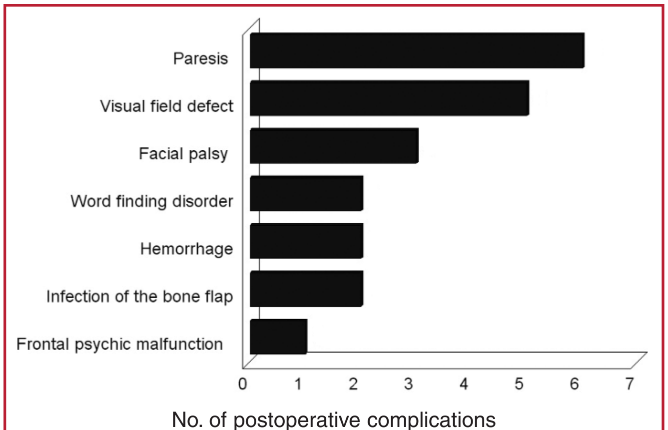

FIGURE 7. *Bar graph showing the types and number of postoperative complications in the entire sample.*

multilobar epilepsy surgery in most previous reports have ranged from 40 to 60% (14, 19, 25). Our results show agreement with the latest published reports. Multilobar epilepsy surgery has the lowest number of patients and the poorest outcome.

Outcome in Relation to Pathological Findings

In well-circumscribed lesions, Engel Class I outcome progressed from 50% to 71.4% after 2 years and from 48.3% to 70.8% after 5 years. In less-well-circumscribed lesions, Engel Class I progressed from 39.3% to 50% after 2 years and from 39.3% to 55.6% after 5 years. There were significant differences in Engel Class I outcome among groups ($P = 0.004$). Previous results show that for patients with less-well-circumscribed lesions, the seizure-free outcome rate is approximately 47 to 72% (1, 9, 17, 29), and for patients with well-circumscribed lesions, the seizure-free outcome rate is in the range of 70 to 80% (10, 13, 28, 29, 31). Our results are in agreement with the newer literature and support the thesis that lesional epilepsy surgery in well-circumscribed lesions has a better outcome.

Duration of Epilepsy and Age of Onset

In our study, we found a statistically and clinically significant correlation between Class I outcome and the duration of epilepsy before surgery ($P = 0.024$). Patients who were operated on within 5 years from epilepsy onset had a lower rate of recurrence compared with patients operated on after more than 5 years. The difference is clear in *Figure 6*, but short duration of epilepsy failed to predict the outcome. The difference between univariate and multivariate analysis indicates that the duration of epilepsy affects the outcome through other factors.

The relationship between good outcome and the duration of presurgical epilepsy is a controversial subject. Some authors have observed that the age at onset, age at time of surgery, and duration of epilepsy were not prognostic factors and therefore had no relationship to the outcome (1, 13, 34). Other authors, however, did find a significant correlation between the shorter preoperative duration of epilepsy and improved outcome. Trevathan and Gilliam (39) reported that early intervention is associated with good outcome in children and adults. Siegel et al. (30) reported good outcome after a second operation, which was related to the short duration of epilepsy. Cohen-Gadol et al. (8) reported a correlation between short duration of epilepsy and good outcome among patients with extratemporal resections. Some authors have reported a relationship between short duration of epilepsy and good outcome with certain abnormalities such as focal cortical dysplasia (17, 29). Our results strongly indicate a relationship between good outcome and short duration of epilepsy among patients with extratemporal epilepsy.

Long-term Outcome

Our results show little and insignificant differences in Engel Class I outcome between follow-up results gathered 2 and 5 years postoperatively. The majority of seizure recurrences in all groups occurred in the first 2 years after surgery. Earlier studies established the first 2-year outcome as a reliable indicator for long-term outcome, but most of these reports dealt with temporal lobe epilepsy surgery (24, 27, 41). Spencer et al. (34) reported no significant difference between temporal and extratemporal resections in the postoperative remission after 2 years of follow-up. Our study shows that the stability of outcome seen 2 years after surgery can be extended to extratemporal epilepsy surgery.

Complications

Our results show a progressive decrease in the complication rate in both permanent and transient morbidity over time, starting in Group I with 8.4% and reaching 1.5% in Group III. Previous reports showed that the complication rates for epilepsy surgery are generally low (3, 4, 6, 22). In 1997, Behrens et al. (3) reported 5.4% as the total rate of neurological complications, with 3.03% causing transient morbidity and 2.33% causing permanent morbidity. Rydenhag and Silander (26) reported minor complications in 8.9% and major complications in 3.1% of patients. Schramm et al. (28) reported transient neurological deficits of 4% and a permanent deficit rate of 2% after frontal lobe epilepsy surgery. Devinsky et al. (10) reported complications in 7.6% of patients.

Patient Selection

The improvement of Group III's outcome reflects the advancement in patient selection as a consequence of accumulated experience gained by the preoperative and operative teams. The third group had the shortest duration of epilepsy before surgery, early adult onset of epilepsy, midlife operation time, the highest number of well-circumscribed lesions (69.6%), and the lowest number of less-well-circumscribed lesions (30.4%). Only 11.1% of the patients had invasive monitoring, compared with 59.3% of the patients in Group I, and all of the patients had lesional epilepsy.

These results intensively point to the role of patient selection in increasing the probability of a good outcome, as some authors have already reported (5, 7, 21, 23). We recommend that patients with extratemporal epilepsy be well selected and referred to presurgical evaluation as early as possible.

CONCLUSION

This study shows the improvement of the outcome of extratemporal epilepsy surgery over time and supports the efficacy of epilepsy surgery to control medically intractable seizures. This development is attributable to appropriate patient selection, accurate presurgical delineation of the epileptic focus, high-resolution MRI, and the experience of the preoperative and surgical teams. This study shows that the outcome of extratemporal epilepsy surgery after 2 years can be used to predict the long-term outcome. The aforementioned information may be important in the decision-making process regarding referral to surgery and tapering of anticonvulsant medications postoperatively.

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COMMENTS

This very large retrospective study reported the results regarding seizure control rates in a consecutive series of more than 200 patients undergoing extratemporal epilepsy surgery between 1991 and 2005. Patients were grouped according to the respective date of surgery in three cohorts. Elsharkawy et al. found that seizure control rates improved over time from about 47% to 60% in this series. Although retrospective in nature, this outcome study provides some important information regarding surgery for extratemporal epilepsy. One strength of the study is the large number of patients. Other important information is the decline in the incidence of Engel Class I seizure-free outcome over 5 years, which is roughly equivalent to a loss of 1% seizure-free patients per year.

Our impression from our own patient groups and from observing the literature over the last 15 years is comparable to the main findings from this report: there seems to be clear improvement in seizure-free outcome with surgery for extratemporal lobe epilepsy. A seizure-free outcome of more than 60% is seen in recent groups using better magnetic resonance imaging technology, whereas 15 or 20 years ago the seizure-free outcome was frequently only approximately 50% (with variations according to various etiologies and locations). This series also confirms multiple findings from our own group and from other authors that patients with neoplasms have a better outcome than those with other causes; the same is true after surgery for temporal lobe epilepsy.

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Medical disciplines are celebrated for their successes, defined by their challenges, and haunted by their failures. Epilepsy surgery, a long and deep partnership between the fields of neurosurgery and neurology, has pockets of progress punctuating a field with little movement. Advances in neuroimaging and neurophysiology/invasive electrode monitoring have expanded the diagnostic arsenal. Novel stimulation and direct-to-brain drug delivery paradigms form an exciting pipeline to future therapies. Yet, it is frustrating to review Penfield's outcomes from 50 years ago and wonder just how far we have come.

This well-documented, consecutive case series between 1991 and 2006 from the Elsharkawy and the Bethel clinic team reported the outcome of extratemporal lobe epilepsy over a 15-year period. They identified a trend that all epilepsy centers would love to see. Looking at three consecutive blocks of 5 years, surgical success consistently improved over time. Although some of the improvement may relate to improved diagnostic and surgical tools, improved patient selection appeared to be the main factor. Their data support this conclusion. Two year postoperative Engel Class I outcome was highest in patients with well-circumscribed lesions such as tumors and vascular lesions (>60%). In contrast, comparable outcomes were lower among patients with focal cortical dysplasia (37%), gliosis (48%), or no pathological lesion (44%). The percentage of patients with tumors and vascular pathological lesions who underwent surgery rose over the past 15 years, whereas those with focal cortical dysplasia, gliosis, and no pathological lesions declined. Further, the sooner epilepsy surgery was performed after the onset of seizures, the better the outcome. The duration of epilepsy among patients undergoing operations declined over the 15-year period.

Unilateral temporal lobe epilepsy and extratemporal lobe epilepsy resulting from localized lesions are the success stories of epilepsy surgery, with seizure-free rates of more than 60%. In highly selected groups (e.g., patients with mesial temporal sclerosis on their nondominant side and concordant seizure onsets), seizure-free rates exceed 80%. Among the estimated 500,000 patients in the United States and 10,000,000 patients worldwide with medically refractory partial epilepsy, less than one-third have unilateral temporal lobe epilepsy or extratemporal epilepsy with well-circumscribed lesions. What about the other two-thirds?

The trend at the Bethel Epilepsy Center to more carefully select candidates for surgery and identify those who have a better chance of achieving seizure freedom parallels the approach at many centers around the world. Improved outcome reflects improved patient selection. But improved outcome for whom? And for the two-thirds who are not offered surgery, is it really better patient selection? Perhaps the goal should be to seek a lower percentage of seizure-free outcomes. Our target population should be widened, away from the "sweet center" of unilateral temporal lobe and extratemporal epilepsy with localized lesions, to include those who are less likely to achieve seizure freedom.

The field of epilepsy surgery focuses on seizure-free outcome as the marker of success. But have we ignored palliation, the opportunity to improve quality of life through a reduction of seizure frequency and medication burden? For the majority of patients with medically refractory partial epilepsy who are "not good candidates for epilepsy surgery," we need to reconsider our definition of a "good candidate." How did we come to decide that seizure freedom was synonymous with surgical success?

Epilepsy surgery shares with cerebral tumor and other neurosurgical procedures a spectrum of disease substrates and a range of respective acceptable outcomes. These outcomes range from permanent alleviation of the disease to palliation or prolongation of the time to recurrence of the disease process. What all of these therapies share is the intention to improve the patient's quality of life. Yet epilepsy surgery, unlike tumor surgery, is perceived as failing unless seizures are completely controlled. Restricting epilepsy to only certain patients, in whom complete success is assured, is analogous to restricting tumor surgery to only benign or low-grade neoplasms. Evaluating success solely on the basis of seizure resolution may deny therapy to patients who may gain worthwhile improvement otherwise. As epilepsy surgery becomes better refined are patients being excluded from consideration?

Who are these patients with refractory partial epilepsy whose seizures are frequent enough to restrict their lives, who have tried multiple antiepileptic drugs and drug combinations with failed results, and who live with ongoing seizures and side effects? For many, epilepsy is a progressive disorder, with deteriorating cognitive and behavioral function and increased rates of suicide and sudden unexplained death. Memory, attention, mental processing speed, executive and language functions, and social skills can all decline as a result of years of seizures pounding on the brain. Depression, anxiety disorders, and psychosis all become more frequent with chronic epilepsy. Perhaps most startling are the rates of sudden unexplained death. Among patients with refractory epilepsy who are considering surgery, the yearly rate of sudden unexplained death in epilepsy is approximately 0.63% (5). Over a decade, that is a 6.3% rate that does not include deaths from other causes. We need to reexamine the benefits as well as the risks of the status quo.

Epilepsy surgery is definitely not for everyone with refractory partial epilepsy. Many patients have multiple or widespread foci that overlap with functional cortex. Surgery could leave a permanent deficit without any benefit. Yet some patients with two or even three seizure foci can benefit when one or two main foci are resected. Seizure frequency and severity may be reduced by 50% or more (2). This is hardly a cure, but surely it is a better place, and, for some, it can be potentially life-saving. Our challenge is not only to increase the frequency of seizure-free outcomes but also to help patients who cannot currently achieve seizure freedom.

The field of epilepsy surgery needs to move beyond the myopic focus on seizures. Yes, well-designed studies have examined deficits in naming and short-term verbal memory after dominant temporal resections (3). Yet, what do we know about real-life executive and social functions after frontal lobe resections? The answer is not much. More recent studies have addressed employment, quality of life, and mood (3, 8, 10). However, we have ignored or avoided questions that Penfield raised a half-century ago concerning the epileptic substrate that he believed caused adverse interictal effects.

The nociferous cortex is defined as dysfunctional epileptogenic tissue that fails to perform its normal functions and impairs the function of other brain areas (4). Penfield and Jasper (7) introduced the concept of nociferous cortex in describing the dramatic behavioral improvements after hemispherectomy in an aggressive boy: "Among patients who have large areas of abnormality in one hemisphere, abnormal behavior may appear, together with advancing mental retardation. The behavioral abnormality is often a more important complaint than the seizures themselves. Radical complete excision may correct the abnormal behavior, stop the seizures, and allow improvement in the patient's mental state" (7, pp 841–842). How do we systematically identify both deficits and improvements after epilepsy surgery? What can we do to predict postoperative cognitive and behavior changes before surgery? Restricted epileptogenic foci can be nociferous. For example, nondominant anterior temporal lobectomy can improve verbal memory (6). Temporal lobectomy can improve metabolic functions ipsilateral and contralateral to the seizure focus, cortically and subcortically (6). Preoperative depression and anxiety often resolve after anterior temporal lobectomy (1, 2).

This article highlighted the capacity of excellent surgical centers to improve outcomes through diagnostic and therapeutic advances as well as more restrictive patient selection. It also highlighted the need to address the problems faced by those patients with epilepsy who do not meet more rigorous selection criteria and raises the question: As epilepsy surgery becomes better refined, are patients who would benefit from surgery being excluded from consideration?

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